

How AI Was Used

I used Claude (Anthropic) as a design and implementation collaborator throughout this assignment — from scoping the design, to writing and testing the code, to a final adversarial pass before submission. This wasn't one-shot generation: I went through the high-level design step by step before any code was written, correcting specific choices along the way (what state actually needed to be tracked, which test cases mattered), and later read the entire codebase and test suite line by line — asking why each specific format, function, or design choice was made, until I could account for every part myself (how `tool_context`, `state`, and `session` relate; why SQLite over Parquet; where retry logic and error handling actually live) rather than just trusting that it worked. A few concrete examples of where pushing back on its first answer led to a real change:

- **Don't trust a green test result — verify it.** Once the implementation looked done, I had Claude critically re-review the whole project against the assignment PDF instead of taking its own word for it. That's what caught a model that had quietly been deprecated by Google mid-project: the evals reported "passing" while every real API call was actually failing — a genuine blind spot in ADK's own eval tooling. Fixing it meant switching models and re-verifying against live calls, not just re-reading test output.
- **Asked harder engineering questions than the assignment required.** Concurrency safety, idempotency under a crash, whether tool calls are authorized before or only inside execution — none explicitly asked for by the PDF. That pressure surfaced two real bugs: a race condition in the transfer logic that could silently drop a concurrent transfer, and a security check that lived only inside one function instead of being independently guarded. Both got fixed and covered by new tests.
- **Unit testing caught a real security gap early on:** an early version of the transfer tool checked "is the session verified" but not "verified *for this specific user*" — a test written to probe exactly that failed, exposing that a verified session could move funds for a different user than the one actually challenged. Fixed before it ever became a demo bug.

A feedback on the assignment

`AgentEvaluator.evaluate()` reports success when every invocation in an eval case errors out (e.g. from a missing key or a dead model), rather than failing — worth reporting upstream, since it can give any candidate false confidence that their evalset is passing when it's never really being exercised at all.